Criterion A: Planning

Defining the problem

For the Internal Assessment, I decided to talk to my primary school Computer Science teacher, Mrs xx, in order to find out if she has any specific needs that I could address. In September 2019 I conducted an interview[1] with Mrs xx, during which it turned out that the provider of my primary school's e-register has recently increased the price of their service and the school was considering switching to a free solution and spending the money saved to partially fund the renovation of their gym. Mrs xx asked me to create a streamlined offline version of the e-register that she would be able to use. She said that my app needs to include the following:

- Login panel
- A way to check attendance
- A calendar, to which tests could be added
- A way to grade test

I agreed with her suggestions. The teacher asked me also to make the program intuitive. During the meeting Mrs xx drew a few sketches to show her design ideas. We agreed to meet again once I'm ready with the project in order to conduct Alpha Testing. If she likes it, she will use it in her classroom and if the app still satisfies her needs, she will extend its functionality by setting up an online server, to which data can be saved and adding functionality such as sending messages and posting announcements.

Rationale for the proposed solution

To make the program easier to write, maintain and extend, I will use classes, such as student, group and test, and separate classes for GUI. Student will have a name and a group; group: tests and students; test: a group, a subject and marks. Each GUI class should provide a blueprint for many similar panels. In order to make my GUI simple and intuitive, I will divide my app into many windows with relatively few buttons.

[1] Appendix A: Interview transcript and sketches made during first the interview

I want the entered data to be as much as possible editable and removable in case of an input error. My program will use file i/o so as to be able to save data on the server in the future. Also, the login details will be encrypted using hash tables to make them unreachable by hackers.

Bearing in mind the above, I decided it would be best to create my app in Java. The reasons for this are as follows:

- I have learned Java at school
- The fact that Java is object-oriented will allow me for an easy creation of classes and objects
- Mrs xx knows Java thus will be able to extend the app
- Java allows for system independence, which is important since some of the computers in my primary school run on Windows while others use Linux
- Java Runtime Environment ensures online security

<u>Stating success criteria</u>

1. In order to use the app, the correct login ("admin") and password ("password") have to be inputted in order to prevent unauthorized use
2. Intuitive interface (few components on each panel, clear labels, using comboBoxes instead of JTextFrames where possible)
3. Ability to add groups and students by typing their names
4. Ability to edit attendance of each student on specific days and periods
5. Ability to add, edit and remove tests by clicking on a button in the calendar and inputting the name of the test, its subject and group
6. Ability to mark tests and view the marks of all groups and students
7. File i/o that allows changes made in one session to be visible in the next one with the encryption of the login and password stored in txt files

Word count: 461